

BEDRU YIMAM AHMED

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RESEARCH INTERESTS

Large Language Models (LLMs), Natural Language Processing for Low-Resource Languages, Machine Learning, Agentic Retrieval-Augmented Generation (RAG), Automatic Speech Recognition (ASR), Multi modal AI, and Applications of NLP to Amharic and other Ethiopian languages.

STATEMENT OF RESEARCH FOCUS

My research goal is to advance natural language processing and large language model methods for low-resource languages, with a particular focus on Amharic and other Ethiopian languages that remain underrepresented in mainstream NLP research. My prior work spans multimodal Amharic fake-news detection, fine-tuning Whisper for Amharic automatic speech recognition, and a survey of ASR challenges across African low-resource languages, and has shaped my interest in techniques such as retrieval-augmented generation (RAG), transfer learning, and data-efficient fine-tuning to make LLMs more effective, reliable, and inclusive for underrepresented languages. I am currently working on mitigating hallucination in Amharic large language models, building on this foundation in low-resource speech and text processing. As a PhD candidate, I intend to extend this work toward scalable, linguistically-grounded methods that bridge the gap between state-of-the-art LLM capabilities and the practical needs of low-resource language communities.

EDUCATION

M.Sc. in Information Technology 2019 – 2020

Bahir Dar Institute of Technology, Bahir Dar University, Ethiopia | CGPA: 3.72/4.00 | Thesis Result: Good

B.Sc. in Information Technology 2014 – 2017

Woldia University, Ethiopia | CGPA: 3.86/4.00

RESEARCH EXPERIENCE & PROJECTS

Mitigating Hallucination in Amharic Large Language Models 2026 – Present

Ongoing independent research – working paper in preparation

- Investigating causes of hallucination in LLM outputs for Amharic, a low-resource language, and exploring mitigation strategies including retrieval-augmented generation (RAG) and grounding techniques.

- Aiming to improve factual reliability of generative models for Amharic-language applications.

Retrieval-Augmented Generation (RAG) Implementation Project 2026

Independent hands-on project | [GitHub – bedr-ux/anc-rag-assistant](https://github.com/bedr-ux/anc-rag-assistant)

- Built a hands-on RAG pipeline as a practical exploration of retrieval-augmented generation techniques, connecting document retrieval with LLM-based generation.
- Used the project to deepen practical understanding of RAG architectures relevant to reducing hallucination and improving factual grounding in low-resource language settings.

Fine-tuning Whisper for Amharic Automatic Speech Recognition (ASR) 2025

Published – arXiv:2503.18485 (see Publications)

- Fine-tuned OpenAI's Whisper ASR model for Amharic using Mozilla Common Voice, FLEURS, and the BDU-speech dataset, producing the Whisper-small-am model.
- Showed that combining FLEURS with new Amharic data outperforms training on new data alone, and that normalizing Amharic homophones meaningfully improves WER and BLEU scores.

Multimodal Amharic Fake News Detection 2024

University-funded research project (Wollo University) – published in Scientific Reports, Oct 2025 (see Publications)

- Designed a multimodal detection pipeline combining NLP and image processing to identify fake news in Amharic-language content, addressing the scarcity of annotated low-resource language datasets.
- Applied deep learning frameworks (TensorFlow, PyTorch) for text and image feature extraction and fusion.

Ethiopian License Plate Detection using Deep CNN 2024

Principal Researcher, University-funded research project (Wollo University) – working paper in preparation

- Developed a convolutional neural network-based computer vision system for automatic detection and recognition of Ethiopian vehicle license plates.

Opal Minerals Classification using Deep Learning (Case of Delanta) 2024

Co-author, University-funded research project (Wollo University) – working paper in preparation

- Built a deep learning classification model to identify opal mineral types using image-based data collected from the Delanta area.

Peer-Review Service – ASTISD-2024 June 2024

- Served as a blind reviewer for full-length paper submissions at the 6th National Annual Research Conference on Advancing Science, Technology and Innovation for Sustainable Development, Kombolcha Institute of Technology, Ethiopia.

PUBLICATIONS

Multimodal Based Amharic Fake News Detection Using CNN and Attention-Based BiLSTM

Alemu Desu Geto, Eshete Derb Emiru, Nuru Seid, Bedru Yimam Ahmed, and 1 more author | *Scientific Reports*, Oct 2025 |

- Presents a multimodal Amharic fake-news detection system built on a 23,856-story dataset collected via Facepacer and the Facebook Graph API. Evaluated multiple CNN/BiLSTM architectures; the CNN with attention-based BiLSTM model achieved 98% accuracy, the best-performing configuration in the study.

Follow this link: [Multimodal based Amharic fake news detection using CNN and attention-based BiLSTM | Scientific Reports](#)

Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions

Sukairaj Hafiz Imam, Babangida Sani, Dawit Ketema Gete, Bedru Yimam Ahmed, and 5 more authors | *Proceedings of the 6th Workshop on African Natural Language Processing (AfricaNLP 2025)*, Jul 2025

- Surveys the major barriers to ASR development for African low-resource languages — data scarcity, linguistic complexity, limited compute, acoustic variability, and bias/privacy concerns — and reviews strategies such as community-driven data collection, self-supervised and multilingual learning, and lightweight model architectures.

Follow this Link: [Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions - ACL Anthology](#)

Whispering in Amharic: Fine-tuning Whisper for Low-resource Language

Dawit Ketema Gete, Bedru Yimam Ahmed, Tadesse Destaw Belay, and 11 more authors | *arXiv:2503.18485 (cs.CL)*, Mar 2025

- Fine-tunes OpenAI's Whisper ASR model for Amharic using Mozilla Common Voice, FLEURS, and the BDU-speech dataset. The resulting Whisper-small-am model improves substantially when fine-tuned on a mix of FLEURS and new Amharic data, and normalizing Amharic homophones is shown to meaningfully improve Word Error Rate (WER) and BLEU scores.

Follow this link: [\[2503.18485\] Whispering in Amharic: Fine-tuning Whisper for Low-resource Language](#)

ACADEMIC & TEACHING EXPERIENCE

Lecturer, Department of Information Technology

Feb 2021 – Present

Wollo University, Kombolcha Institute of Technology, Ethiopia

- Teach a broad range of IT and computer science courses, including programming languages (Python, Java, PHP, C++, JavaScript, SQL), data structures, networking, artificial intelligence, blockchain, and IoT.
- Mentor and advise 80+ students on career-aligned skill development, project implementation, and technical documentation.
- Supervise student projects and research, and contribute to curriculum development aligned with emerging technologies.
- Developed a Student Information System (registration-to-grade reporting) using PHP, JavaScript, and SQL, and used system analytics to identify at-risk students for early intervention.
- Implemented progress-tracking features (HTML/CSS/JS) for an online e-learning platform at Kombolcha RADA College to monitor student engagement.

Assistant Lecturer / Graduate Assistant II

2017 – 2019

Wollo University, Kombolcha Institute of Technology, Ethiopia

- Supported course delivery and lab sessions in the Department of Information Technology prior to MSc study leave (2019–2021).

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Certificate of Completion, “Teaching in an Ethiopian University of Applied Sciences” Professional Development Program – jointly offered by Wollo University and Arizona State University, March 2025.
- Higher Diploma in Teaching in Higher Education – Wollo University Institute of Teachers' Education and Behavioural Sciences.
- National Qualification Certificate (Level IV), Web Development and Database Administration – Ethiopian Ministry of Labor and Skills, 2024 (valid through 2027).

TECHNICAL SKILLS

- LLM Engineering: Fine-tuning LLMs, LoRA, QLoRA, LangChain, LangGraph, Retrieval-Augmented Generation (RAG)
- Data Science & Machine Learning: Data analysis, model evaluation, statistical methods, feature engineering
- AI / ML: TensorFlow, PyTorch, NLP, Computer Vision, NumPy, Pandas
- Programming: Python, Java, PHP, JavaScript, SQL, C++
- Web Development: HTML/CSS

LANGUAGES

Amharic – Native English – Advanced (professional working proficiency)